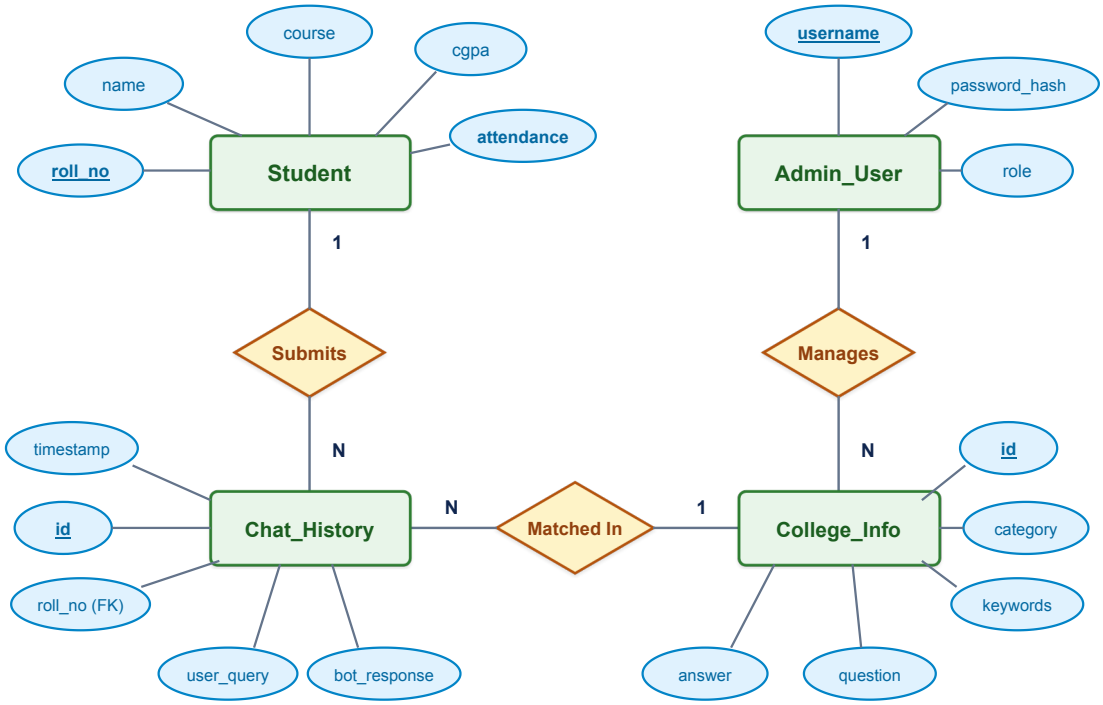
AI CHATBOT FOR COLLEGE

ACADEMIC UML & ARCHITECTURAL SPECIFICATION

✓ 4 Entities (Student, Admin, Chat, Info) ✓ Classic Notation (Ovals, Diamonds) ✓ 3NF Normalization Defense

Entity-Relationship Diagram (Classic Notation) — 4 Core Entities & Relations

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

Entity (Table)

Attribute (Column) — id = Primary Key

Relationship

Link / Cardinality (1:N)

Fig: Entity-Relationship Diagram (ERD — Classic Notation) — 4 core entities (Student, Admin_User, Chat_History, College_Info) with attributes (PK underlined, FK indicated), cardinalities (1:N, N:1), and official symbol key matching faculty review specifications.